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8. *(a)* (i) Define the term *standard enthalpy change of combustion*, $\Delta_c H^\theta$. [2]

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(ii) The standard enthalpy change of combustion of ethane is –1561 kJ mol^{-1}.

The values of some average bond enthalpies are shown in the table below.

Bond	Average bond enthalpy / kJ mol^{-1}
C—C	348
O═O	495
C═O	799
O—H	463

$$C_2H_6 \ + \ .... \ O_2 \longrightarrow 2CO_2 \ + \ 3H_2O$$

Balance the equation for the combustion of ethane and use the information given to calculate the average bond enthalpy for a C—H bond. [4]

Bond enthalpy = ... kJ mol^{-1}

(iii) Suggest a reason why bond enthalpies are described as being **average** bond enthalpies. [1]

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(b) Charcoal consists mainly of carbon. It has been produced for many centuries by heating wood in the absence of oxygen. Natural gas consists mainly of methane and is obtained from underground sources. It was formed in a similar way to coal and oil.

The enthalpy changes of combustion of carbon and methane are in the table.

Substance	Enthalpy change of combustion / kJ mol^{-1}
carbon, C	–394
methane, CH_4	–889

Two students were discussing the use of charcoal and methane as fuels.

One said that 'methane produced more heat per gram when burned so that it was a better fuel'.

The other student said that 'the use of charcoal contributed less to the overall increase of carbon dioxide levels in the atmosphere'.

Discuss these two statements. [6 QER]

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